Advantages of Agile Methodology

Following are the advantages of agile methodology:

1. **Focus on Customer Value:** Agile places a high priority on providing customers with value by attending to their requirements and preferences. Agile guarantees that the most important features are produced first and that iterative changes are driven by customer feedback by dividing work down into small, manageable tasks.
2. **Enhanced Team Morale and Motivation:** Agile gives teams the freedom to own their work and decide together. Team members feel motivated, proud, and owned when they have this autonomy together with a focus on providing value and ongoing growth.
3. **Stakeholder Collaboration:** Throughout the development process, agile promotes strong coordination between product owners, developers, and other stakeholders. Better communication, a common understanding of the objectives, and ongoing feedback are all fostered by this partnership, which produces results that are higher quality and boost stakeholder satisfaction.
4. **Early and Continuous Delivery:** Agile encourages the tiny, incremental releases of functional software. This gives early access to observable progress and facilitates early input and validation for stakeholders. Continuous delivery reduces risks by spotting problems early on and taking action to fix them.
5. **Delivering high-quality software:** It is a key component of agile development, and this is emphasized by techniques like continuous integration, automated testing, and frequent inspection and modification. Agile guarantees that the software satisfies the required standards and lowers the likelihood of faults by integrating quality assurance throughout the development process.

Disadvantages of the Agile Methodology

Following are the disadvantages of the agile methodology:

1. **Lack of Predictability:** Project timeframes and outcomes might be difficult to predict with accuracy due to Agile iterative and incremental methodology. Stakeholders who need set budgets or timeframes may find this unpredictability troublesome.
2. **Dependency on Customer Availability:** Agile highly depends on ongoing customer and stakeholder feedback and participation. Customers who are unavailable or who don't know enough about the domain can impede development and slow it down.
3. **Scaling Agile**: While Agile works effectively for small to medium-sized teams working on relatively basic projects, scaling Agile methods to bigger teams or more complicated projects can be more difficult. As the project grows, it gets harder to maintain coordination, alignment, and communication.
4. **Dependency on Team Dynamics:** Agile's focus on self-organizing, cross-functional teams with the authority to reach decisions together is paramount. Inadequate communication within the team or a lack of experience or expertise among team members can negatively affect output quality and productivity.

5. **Increased Overhead:** Planning, coordinating, and communicating take more time and effort when using agile frameworks like Scrum. This overhead can take a lot of time, especially for projects with short deadlines or small teams.

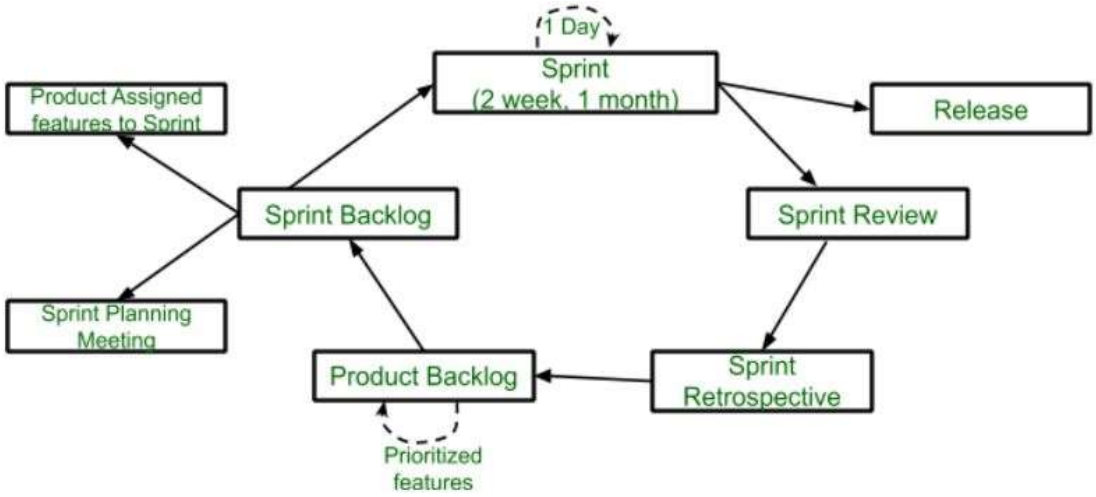
What is Scrum in Software Development?

Scrum is a management framework that teams use to self-organize tasks and work towards a common goal. It is a framework within which people can address complex adaptive problems while the productivity and creativity of delivering products are at the highest possible value. Scrum is a management framework that teams use to self-organize and work towards a common goal.

- Scrum allows us to develop products of the highest value while making sure that we maintain creativity and productivity.
- The iterative and incremental approach used in scrum allows the teams to adapt to the changing requirements.

Silent features of Scrum

- Scrum is a light-weighted framework
- Scrum emphasizes self-organization
- Scrum is simple to understand
- Scrum framework helps the team to work together
- Lifecycle of Scrum



- **Sprisnt:** A Sprint is a time box of one month or less. A new Sprint starts immediately after the completion of the previous Sprint. **Release:** When the product is completed, it goes to the Release stage.

What is a sprint cycle?

- A sprint cycle is a series of iterations that break down a project into smaller, more manageable pieces
- Sprints are typically two to four weeks long, but can vary depending on the project
- A new sprint begins immediately after the previous sprint ends

- **Sprint Review:** If the product still has some non-achievable features, it will be checked in this stage and then passed to the Sprint Retrospective stage.
- **Sprint Retrospective:** In this stage quality or status of the product is checked. **Product Backlog:** According to the prioritize features the product is organized.
- **Sprint Backlog:** Sprint Backlog is divided into two parts Product assigned features to sprint and Sprint planning meeting.